

第二章：均匀系的平衡性质

麦克斯韦关系

1. 麦克斯韦关系是热力学基本微分方程的直接结果。

2. $P-T-V$ 系统。

热一定律

$$dU = \delta Q + \delta W$$

膨胀功

$$\delta W = -P dV$$

热二定律, 可逆

$$\delta Q = T dS$$

$$dU = T dS - P dV$$

包括了热一、热二定律, 微小可逆过程。

看作以 (S, V) 为独立变量, $U = U(S, V)$

$$dU = \left(\frac{\partial U}{\partial S}\right)_V dS + \left(\frac{\partial U}{\partial V}\right)_S dV$$

比较可得

$$\left\{ \begin{array}{l} \left(\frac{\partial U}{\partial S}\right)_V = T \\ \left(\frac{\partial U}{\partial V}\right)_S = -P \end{array} \right\} \text{麦克斯韦关系.}$$

因 $T > 0$, $(\frac{\partial U}{\partial S})_V > 0$, V 不变时, U 与 S 的变化倾向相同.

因 $P > 0$, $(\frac{\partial U}{\partial V})_S < 0$, S 不变时, U 与 V 变化倾向相反.

U 是态函数, dU 是全微分.

$$\frac{\partial^2 U}{\partial V \partial S} = \frac{\partial^2 U}{\partial S \partial V} \quad \text{与次序无关.}$$

$$\left(\frac{\partial T}{\partial V}\right)_S = \frac{\partial^2 U}{\partial S \partial V} = \frac{\partial^2 U}{\partial V \partial S} = -\left(\frac{\partial P}{\partial S}\right)_V$$

2. 勒让德变换.

$$x dy \equiv d(xy) - y dx$$

将 $x dy$ 变为 $-y dx$.

$$dU = T dS - P dV, \quad T \leftrightarrow S, P \leftrightarrow V, \text{ 共轭变量.}$$

$$T dS \equiv d(TS) - S dT$$

$$dU = d(TS) - S dT - P dV$$

$$d(U - TS) = dF = -S dT - P dV$$

$$F = F(T, V)$$

$$\begin{cases} S = -\left(\frac{\partial F}{\partial T}\right)_V \\ P = -\left(\frac{\partial F}{\partial V}\right)_T \end{cases}$$

如果知道 $F = F(T, V)$, 求微商可得熵
与物态方程. $P = -\left(\frac{\partial F}{\partial V}\right)_T = f(T, V)$

$$\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial P}{\partial T}\right)_V$$

自然变量 (S, V) , (T, V) , 互相变换.

类似的.

$$(S, V) \rightarrow (S, P) \quad H = H(S, P)$$

$$PdV = d(pV) - VdP$$

$$dU = TdS - PdV$$

$$d(U + PV) = dH = TdS + VdP$$

$$\left(\frac{\partial H}{\partial S}\right)_P = T \quad \left(\frac{\partial H}{\partial P}\right)_S = V$$

$$\left(\frac{\partial T}{\partial P}\right)_S = \frac{\partial^2 H}{\partial S \partial P} = \left(\frac{\partial V}{\partial S}\right)_P$$

$$(S, V) \rightarrow (T, P)$$

$$TdS = d(TS) - SdT$$

$$d(U + PV - TS) = dG = -SdT + VdP$$

$$\left(\frac{\partial G}{\partial T}\right)_P = -S \quad \left(\frac{\partial G}{\partial P}\right)_T = V$$

$$-\left(\frac{\partial S}{\partial P}\right)_T = \left(\frac{\partial V}{\partial T}\right)_P$$

如知道了 $G(T, P)$, 则可求微商得温度和物态方程, $V = \left(\frac{\partial G}{\partial P}\right)_T = f(T, P)$

还可以有更多的麦克斯韦关系. 例如:

$$dU = TdS - PdV \rightarrow dS = \frac{1}{T}dU + \frac{P}{T}dV$$

$$S = S(U, V)$$

3. 应用例子.

例1. $C_p - C_v$

$$C_v = \left(\frac{dQ}{dT}\right)_v \quad dQ = TdS \quad C_v = T\left(\frac{\partial S}{\partial T}\right)_v$$

$$C_p = \left(\frac{dQ}{dT}\right)_p \quad dQ = TdS \quad C_p = T\left(\frac{\partial S}{\partial T}\right)_p$$

$$S = S(T, V, p) = S(T, V(T, p))$$

$$\left(\frac{\partial S}{\partial T}\right)_p = \left(\frac{\partial S}{\partial T}\right)_V + \left(\frac{\partial S}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_p$$

$$T\left(\frac{\partial S}{\partial T}\right)_p - T\left(\frac{\partial S}{\partial T}\right)_V = T\left(\frac{\partial S}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_p$$

$$C_p - C_V = T\left(\frac{\partial S}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_p$$

利用麦克斯韦关系 $\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial p}{\partial T}\right)_V$

$$C_p - C_V = T \left(\frac{\partial p}{\partial T}\right)_V \left(\frac{\partial V}{\partial T}\right)_p \quad ①$$

$C_p - C_V$ 可由物态方程求出 $\begin{cases} p = f(T, V) \\ V = f(T, p) \end{cases}$

利用 $\alpha = \frac{1}{V} \left(\frac{\partial V}{\partial T}\right)_p$ $\beta = \frac{1}{p} \left(\frac{\partial p}{\partial T}\right)_V$

$\kappa_T = -\frac{1}{V} \left(\frac{\partial V}{\partial p}\right)_T$ $\alpha = \kappa_T \beta p$

$$C_p - C_V = T (\beta p) (\alpha V) = T \frac{\alpha}{\kappa_T} \alpha V = TV \frac{\alpha^2}{\kappa_T}$$

$\kappa_T > 0$ $C_p > C_V$

如果系统为理想气体 $pV = NkT$, 由①

$$C_p - C_V = T \cdot \frac{Nk}{V} \cdot \frac{Nk}{p} = Nk$$

例2. $\left(\frac{\partial U}{\partial V}\right)_T$

由 $dU = TdS - PdV$, 选 (V, T) 作变量.

$$S = S(V, T) \quad dS = \left(\frac{\partial S}{\partial V}\right)_T dV + \left(\frac{\partial S}{\partial T}\right)_V dT$$

$$dU = T \left[\left(\frac{\partial S}{\partial V}\right)_T dV + \left(\frac{\partial S}{\partial T}\right)_V dT \right] - PdV$$

$$= \left(T \left(\frac{\partial S}{\partial V}\right)_T - P \right) dV + \left(\frac{\partial S}{\partial T}\right)_V dT$$

$$\left(\frac{\partial U}{\partial V}\right)_T = T \left(\frac{\partial S}{\partial V}\right)_T - P = T \left(\frac{\partial P}{\partial T}\right)_V - P$$

如系统为理想气体 $PV = NKT$

$$\left(\frac{\partial P}{\partial T}\right)_V = \frac{NR}{V} = \frac{P}{T} \Rightarrow \left(\frac{\partial U}{\partial V}\right)_T = 0$$

理想气体内能与 V 无关, 只是 T 的函数.

4. 可测量量 / 不可测量量.

可测量量 $\left\{ \begin{array}{l} \text{① 状态变量} \\ \text{② 各种热容} \\ \text{③ 与物态过程相联系的数量} \end{array} \right\}$ 响应函数.

不可测量量 = $U, S, H, F, G, \left(\frac{\partial S}{\partial V}\right)_T$ 等.